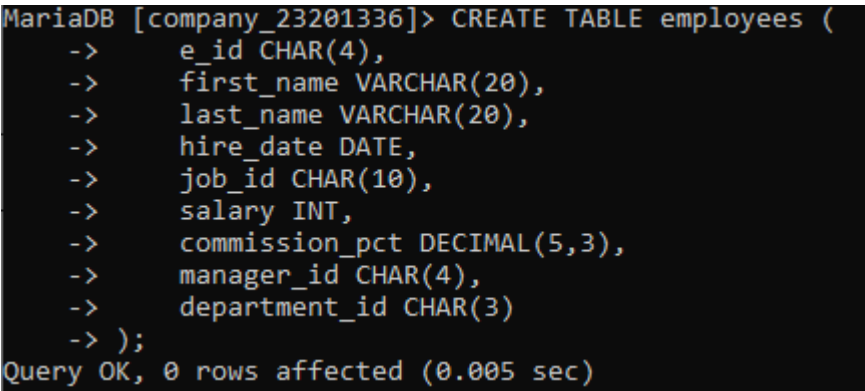


CSE370 : Database Systems

Assignment 01/02/03 | Spring 2026

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No 1 Query (as Plain Text)	CREATE TABLE employees (e_id CHAR(4), first_name VARCHAR(20), last_name VARCHAR(20), hire_date DATE, job_id CHAR(10), salary INT, commission_pct DECIMAL(5,3), manager_id CHAR(4), department_id CHAR(3));
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	INSERT INTO employees VALUES ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'),

	('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
No 2 SS (of Query & Output in Shell)	<pre> MariaDB [company_23201336]> INSERT INTO employees VALUES -> ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), -> ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), -> ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), -> ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), -> ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), -> ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), -> ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), -> ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), -> ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), -> ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), -> ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), -> ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), -> ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), -> ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), -> ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04'); Query OK, 15 rows affected (0.027 sec) Records: 15 Duplicates: 0 Warnings: 0 </pre>

No 3 Query (as Plain Text)	SELECT first_name, last_name, hire_date, department_id FROM employees WHERE hire_date = (SELECT MAX(hire_date) FROM employees);
No 3 SS (of Query & Output in Shell)	<pre> MariaDB [company_23201336]> SELECT first_name, last_name, hire_date, department_id -> FROM employees -> WHERE hire_date = (SELECT MAX(hire_date) FROM employees); +-----+-----+-----+-----+ first_name last_name hire_date department_id +-----+-----+-----+-----+ Kevin Lewis 2022-03-12 D04 +-----+-----+-----+-----+ 1 row in set (0.001 sec) </pre>
No 4 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id FROM employees e WHERE salary = (SELECT MAX(salary) FROM employees WHERE department_id = e.department_id);

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [company_23201336]> SELECT first_name, last_name, e_id, salary, department_id
-> FROM employees e
-> WHERE salary = (
-> SELECT MAX(salary)
-> FROM employees
-> WHERE department_id = e.department_id
-> );
```

first_name	last_name	e_id	salary	department_id
Patricia	Garcia	E010	85000	D01
Thomas	Lee	E011	80000	D02
Maria	White	E012	75000	D03
Daniel	Harris	E013	78000	D04

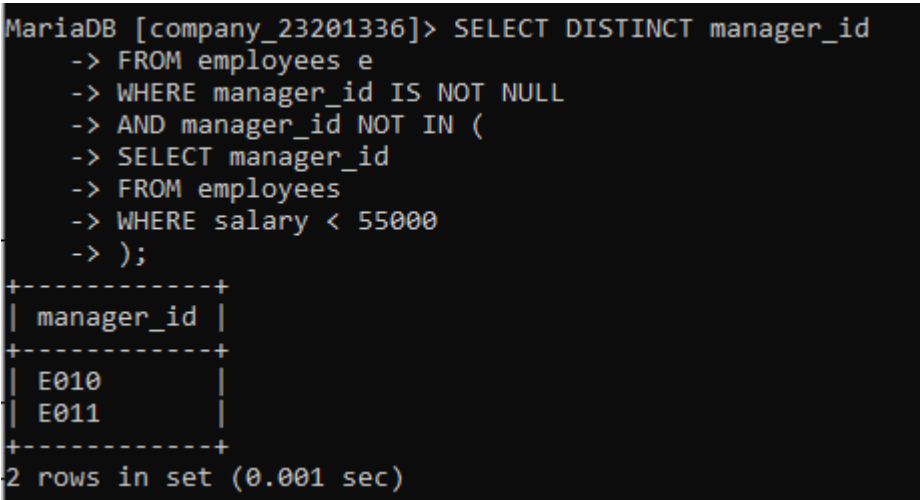
```
4 rows in set (0.000 sec)
```

No 5 Query
(as Plain Text)

```
SELECT first_name, last_name, e_id, commission_pct, department_id
FROM employees
WHERE department_id='D02'and commission_pct> all(select commission_pct
FROM employees
WHERE department_id='D01' and commission_pct is not null);
```

No 5 SS (of Query & Output in Shell)	<pre> MariaDB [company_23201336]> SELECT first_name, last_name, e_id, commission_pct, department_id -> FROM employees -> WHERE department_id='D02'and commission_pct> all(select commission_pct -> FROM employees -> WHERE department_id='D01' and commission_pct is not null); +-----+-----+-----+-----+-----+ first_name last_name e_id commission_pct department_id +-----+-----+-----+-----+-----+ Jane Smith E002 0.150 D02 Emily Davis E004 0.200 D02 James Taylor E007 0.180 D02 Thomas Lee E011 0.100 D02 +-----+-----+-----+-----+-----+ 4 rows in set (0.000 sec) </pre>
No 6 Query (as Plain Text)	<pre> SELECT department_id, COUNT(*) AS total_employees FROM employees GROUP BY department_id HAVING MAX(salary) <= 70000; </pre>
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [company_23201336]> SELECT department_id, COUNT(*) AS total_employees -> FROM employees -> GROUP BY department_id -> HAVING MAX(salary) <= 70000; Empty set (0.000 sec) </pre>
No 7 Query (as Plain Text)	<pre> SELECT department_id, job_id, salary FROM employees </pre>

	WHERE salary < any(select salary FROM employees WHERE department_id= employees. department_id);
No 7 SS (of Query & Output in Shell)	<pre> MariaDB [company_23201336]> SELECT department_id, job_id, salary -> FROM employees -> WHERE salary < any(select salary -> FROM employees -> WHERE department_id= employees. department_id); +-----+-----+-----+ department_id job_id salary +-----+-----+-----+ D01 IT_PROG 65000 D02 SA_REP 55000 D01 IT_PROG 62000 D02 SA_REP 58000 D01 IT_PROG 60000 D03 HR_REP 50000 D02 SA_REP 60000 D04 FI_ACC 52000 D01 IT_PROG 67000 D02 SA_MAN 80000 D03 HR_MAN 75000 D04 FI_MAN 78000 D03 HR_REP 48000 D04 FI_ACC 51000 +-----+-----+-----+ 14 rows in set (0.000 sec) </pre>
No 8 Query (as Plain Text)	SELECT DISTINCT manager_id FROM employees e WHERE manager_id IS NOT NULL

	<pre> AND manager_id NOT IN (SELECT manager_id FROM employees WHERE salary < 55000); </pre>
No 8 SS (of Query & Output in Shell)	 <pre> MariaDB [company_23201336]> SELECT DISTINCT manager_id -> FROM employees e -> WHERE manager_id IS NOT NULL -> AND manager_id NOT IN (-> SELECT manager_id -> FROM employees -> WHERE salary < 55000 ->); +-----+ manager_id +-----+ E010 E011 +-----+ 2 rows in set (0.001 sec) </pre>
No 9 Query (as Plain Text)	<pre> SELECT first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e WHERE commission_pct = (SELECT MAX(commission_pct) FROM employees WHERE manager_id = e.manager_id); </pre>

No 9 SS
(of Query & Output
in Shell)

```
MariaDB [company_23201336]> SELECT first_name, last_name, e_id, salary, department_id, commission_pct  
-> FROM employees e  
-> WHERE commission_pct = (  
-> SELECT MAX(commission_pct)  
-> FROM employees  
-> WHERE manager_id = e.manager_id  
-> );
```

first_name	last_name	e_id	salary	department_id	commission_pct
Emily	Davis	E004	58000	D02	0.200
Sarah	Brown	E006	50000	D03	0.150

```
2 rows in set (0.000 sec)
```

No 10 Query
(as Plain Text)

```
SELECT job_id, AVG(salary) AS avg_salary  
FROM employees  
GROUP BY job_id  
HAVING AVG(salary) < (  
SELECT AVG(salary) FROM employees  
);
```


No 10 SS
(of Query & Output
in Shell)

```
MariaDB [company_23201336]> SELECT job_id, AVG(salary) AS avg_salary
-> FROM employees
-> GROUP BY job_id
-> HAVING AVG(salary) < (
-> SELECT AVG(salary) FROM employees
-> );
```

job_id	avg_salary
FI_ACC	51500.0000
HR_REP	49000.0000
SA_REP	57666.6667

3 rows in set (0.000 sec)